

Aashish Gupta

VICO Postdoctoral Fellow at the Department of Astronomy, University of Virginia, USA

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RESEARCH INTERESTS

I study how stars and planets form, primarily using observations from a range of telescopes. My recent work focuses on the interactions between star-forming environments and planet-forming disks, and on how they alter the material available to build planets.

HIGHLIGHTS

- 16 (+8 submitted) refereed research publications, including 8 (+2 submitted) with major contributions ([more details](#)).
- PI on 8 approved programs across ALMA (105.8 h), GBT (21.4 h), SMA (8 h), and VLT (3.7 h) ([more details](#)).
- 28 research talks given, including 14 invited talks ([more details](#)).
- Mentored 6 students on astronomy-related projects, ranging from a high school student to graduate students ([more details](#)).
- Founded and running “The Big Science Show”, a monthly public science outreach event in Charlottesville with 150+ subscribers.
- Conceived and chairing the organizing committee for the “DISCO: Disks in Context”, a star and planet formation conference with 60+ participants.

EDUCATION

Doctor of Philosophy (Physics) European Southern Observatory (ESO) and Ludwig Maximilian University (LMU), Germany	<i>2021 – 2024</i> <i>Magna cum laude</i>
Master of Science (Astronomy) National Central University (NCU), Taiwan	<i>2019 – 2021</i>
Bachelor of Technology (Mechanical Engineering) Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur, India	<i>2015 – 2019</i>

PROFESSIONAL APPOINTMENTS

<i>(Upcoming)</i> Lecturer , University of Virginia (UVA), USA	<i>Fall 2026</i>
VICO postdoctoral fellow , University of Virginia (UVA), USA <i>Advisor: Dr. Ilse Cleaves</i> I am leading multiple research projects to characterize the impact of star-forming environments on planet-forming disks.	<i>2025 – present</i>
Doctoral researcher , European Southern Observatory (ESO), Germany <i>Advisor: Dr. Anna Miotello</i> I demonstrated that planet-forming disks often interact with surrounding molecular clouds, and such clouds can drastically enhance their mass-budget to form planets.	<i>2021 – 2024</i>
Graduate researcher , National Central University (NCU), Taiwan <i>Advisor: Prof. Chen, Wen-Ping</i> I diagnosed the distribution of the young stars with respect to their natal molecular clouds.	<i>2019 – 2021</i>

- Summer intern**, Academia Sinica Institute of Astronomy and Astrophysics (ASIAA), Taiwan 2020
Advisor: Dr. Yen, Hsi-Wei
 I investigated influence of magnetic fields on gas kinematics within protostellar envelopes.
- Project-based internship**, Aryabhata Research Institute of Observational Sciences (ARIES), India 2018
Advisor: Dr. Jeewan Pandey
 I developed a pipeline to detect and analyze stellar flares.
- Summer intern**, Aryabhata Research Institute of Observational Sciences (ARIES), India 2017
Advisor: Dr. Sneh Lata
 I developed a pipeline to analyse time series data of RR Lyrae Variables.
- Undergraduate researcher**, Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur, India 2016
Advisor: Prof. Puneet Tandon
 I assisted in the design and fabrication of a 3D printing machine.

SKILLS

- Programming** Python, C/C++, Octave, HTML, Shell scripting, L^AT_EX
- Languages** Hindi (Native), English (Proficient, IELTS score 8), German (Basic, A1), Spanish (Basic, A1), Mandarin Chinese (Basic speaking and listening)

OBSERVING EXPERIENCE

PI Projects:

1. ALMA project “The invisible majority: Testing CI as a tracer for CO-dark streamers” (6.4 h on 12 m array, 24.9 h on ACA) 2026.1.00679.S
2. ALMA project “Zooming Out and Peering In: The First Statistical Study of Infall onto Class II Disks” (12.3 h on 12 m array) 2025.1.00328.S
3. ALMA project “Paradigm Shift or Rare Events: Testing the Frequency of Late Infall Using Herbig Disks” (16.2 h on ACA, 25.3 h on TP) 2025.1.00197.S
4. ALMA project “Is infall of material misaligning planet-forming disks?” (8.7 h on 12 m array) 2025.1.00803.S
5. ALMA project “Zooming out: Disentangling the origins of large-scale structures around planet-forming disks” (11.8 h on ACA) 2024.1.00524.S
6. GBT project “Characterizing gas reservoirs feeding planet-forming disks” (21.4 h) GBT26A-579
7. SMA project “In or Out: Spectral-Line Survey of a Streamer” (8 h) 2025A-S031
8. VLT project “Exploring effects of infall of material onto Class II systems” (3.7 h on X-Shooter) 113.26GY

Involvement in Large Programs:

1. Disk-Exoplanet C/Onnection (DEC/O) 2022.1.00875.L
 138.8 h on ALMA, PI: Ilse Cleaves
 I am leading the efforts on characterizing the large-scale structures observed in DEC/O. I was also involved in the development of imaging pipeline, submitting follow-up proposals (2 accepted), and giving feedback on manuscripts.
2. Chemistry of Herbig Environments and their Exoplanet Relationships (CHEER) 2024.1.01001.L

89.7h on ALMA, PI: Jamila Pegues

I am co-leading the efforts on characterizing the large-scale structures observed in CHEER. I am also involved in the development of imaging pipeline and submitting follow-up proposals (2 accepted).

3. The JCMT Transient Survey *M16AL001*
200+h on JCMT, PIs: Gregory Herczeg, Doug Johnstone

I lead the analysis of coadd image for the Ophiuchus region. I also helped with literature review of interesting sources.

CoI Projects:

CoI on 14 approved programs (700+h): 13 ALMA, 2 JWST, 2 VLT, 2 APEX, 1 SMA, and 1 SMT.

Hands-on Observing:

Assisted with optical observations at ARIES, India and Lulin Observatory, Taiwan.

SOFTWARE DEVELOPMENT

TIPSY – Trajectory of Infalling Particles in Streamers around Young stars *2024*

I developed TIPSY, the first-of-its-kind open-source code for characterizing “streamers”, i.e., trails of gas falling onto young stellar systems. TIPSY fits the molecular-line observations of streamers in 3D position–position–velocity space and constrains dynamical parameters like energy, angular momenta, infall timescale, etc. TIPSY-based analysis has been central to over 8 refereed publications so far (visit the [GitHub repository](#) for more details).

MENTORING EXPERIENCE

- Supervising Mason Mabry for Interconnected Cosmos Undergraduate Research Program at UVA *2026 – present*
- Co-supervising Noshin Yesmin for her Ph.D. project at UVA *2026 – present*
- Supervised Andres Zuleta for “Cool Frontiers: Exploring Dust and Ice in the Cosmos” at UVA *2025*
- Co-supervised Lauren Mason for the ESO Summer Research Programme *2024*
- Supervised Jayden Burg, a high school student, for one week internship (Schülerpraktikum) *2024*
- Co-supervised Alessandro Ruzza for the ESO Summer Research Programme *2023*
- Student guide for the first year students at IIITDMJ *2016*

TEACHING EXPERIENCE

- (Upcoming)* ASTR 3150 “The Interstellar Medium: From Hydrogen to Humans” *Fall 2026*
- Lecture on planet formation for ASTR 1610: “Introduction to Astronomical Research” at UVA *Spring 2026*
- Academic Helper for Physics and Mathematics at IIITDMJ Counseling Services *2016 – 2018*

ACADEMIC SERVICE

- Chair of the organizing committee for the “DISCO: Disks in Context”, a star and planet formation conference *2025 – present*
- Chair of Community Building Committee of the UVA Postdoc Association *2025 – present*
- Local Organizing Committee member for “Cool Frontiers: Exploring Dust and Ice in the Cosmos” at UVA *2025*
- Local Organizing Committee member for DECO All Team meeting at ESO, Germany *2024*

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| 5. Local Organizing Committee member for “New Heights in Planet Formation” at ESO, Germany | 2024 |
| 6. Scientific Assistant for ESO Observing Programmes Committee | 2024 |
| 7. Member of selection committee for ESO Summer Research Programme | 2024 |
| 8. Co-organizer of Wine & Cheese seminars at ESO, Germany | 2023 – 2024 |
| 9. Chaired for Hypatia Colloquium at ESO, Germany | 2023 |
| 10. Refereed for Science Advances and The Astrophysical Journal | |

OUTREACH

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| 1. Founding organizer for “The Big Science Show”. Every month, we host two local scientists, from any field of research, to present their topic of interest to the community. | 2026 – present |
| 2. Co-organizer of “Go Explore the Universe” summer camp at Charlottesville, USA | 2026 |
| 3. Outreach talk at Astronomy on Tap, Charlottesville, USA | 2026 |
| 4. Public night talk at McCormick Observatory at Charlottesville, USA | 2026 |
| 5. Co-organizer of “Girls Exploring the Universe” summer camp at Charlottesville, USA | 2025 |
| 6. Co-organizer of “Astronomy for Non-astronomers” monthly talk series at ESO, Germany | 2024 |
| 7. Coordinator of The Astronomy and Physics Society of IIITDM Jabalpur, India. We organized events like telescope nights, exhibitions, quizzes, etc. to promote astronomy among a variety of audiences. | 2016 – 2019 |
| 8. E-Astronomer at RAD@home Astronomy Collaboratory, an Indian citizen-science research programme | 2018 – 2019 |

AWARDS AND GRANTS

Total awards and grants worth ~\$441,730.

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| 1. \$ 218,000 – VICO postdoctoral fellowship at UVA, USA | 2025 – 2028 |
| 2. \$ 6,500 – Global Initiative Grant from Center for Global Inquiry and Innovation at UVA, USA for organizing “DISCO: Disks in Context” conference | 2026 |
| 3. € 172,550 (~\$ 200,000) – DFG funding for Ph.D. at ESO, Germany | 2021 – 2024 |
| 4. € 4,000 (~\$ 4,500) – ERC DUSTBUSTERS funding for a two-month secondment at the University of Hawaii, USA | 2022 |
| 5. NT\$ 361,100 (~\$ 11,180) – MOST scholarship (tuition fees waiver and support of living expenses) for master’s program at NCU, Taiwan | 2019 – 2021 |
| 6. NT\$ 20,000 (~\$ 620) – Award for academic performance at NCU, Taiwan | 2020 |
| 7. Poster presentation award, Annual Meeting of the Physical Society of Taiwan | 2020 |
| 8. ₹ 60,000 (~\$ 630) – ARIES funding for semester-long internship, Nainital, India | 2018 |
| 9. Certificate of Merit for academic excellence at IIITDM Jabalpur, India | 2017 |

PARTICIPATION IN MEETINGS

Invited talks:

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| 1. Astronomy Department Talk, University of Michigan, Ann Arbor, USA | July 2026 |
| 2. Astronomy Seminar, Tata Institute of Fundamental Research, Mumbai, India | February 2026 |
| 3. Astronomy Seminar, University of Connecticut, Storrs, USA | October 2025 |
| 4. Star and Exoplanet Seminar, Yale University, New Haven, USA | October 2025 |
| 5. Center for Astrophysics Harvard & Smithsonian, Cambridge, USA | June 2025 |

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| 6. European Astronomical Society Annual Meeting (EAS), Cork, Ireland | <i>June 2025</i> |
| 7. StarPlan seminar, University of Copenhagen, Copenhagen, Denmark | <i>December 2024</i> |
| 8. Astronomy seminar, Universidad Diego Portales, Santiago, Chile | <i>September 2024</i> |
| 9. Astronomy seminar, University of Exeter, Exeter, UK | <i>February 2024</i> |
| 10. Core2disk III workshop, Paris, France | <i>October 2023</i> |
| 11. Origins seminar at Chalmers University, Gothenburg, Sweden | <i>August 2023</i> |
| 12. ESO star and planet formation seminar, Garching, Germany | <i>February 2023</i> |
| 13. ECOGAL seminar, Online | <i>February 2023</i> |
| 14. MPE-CAS journal club on star and planet formation, Garching, Germany | <i>July 2022</i> |

Contributed talks:

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| 1. Emerging Researchers in Exoplanet Science Symposium XI, Ann Arbor, Michigan | <i>July 2026</i> |
| 2. MIAPbP workshop on “Planet Formation across various Environments and Epochs”, Garching, Germany | <i>May 2026</i> |
| 3. ESO Wine & Cheese Seminar | <i>May 2026</i> |
| 4. UVA Postdoc Research Symposium | <i>May 2026</i> |
| 5. VICO Fall Workshop 2025, Charlottesville, USA | <i>December 2025</i> |
| 6. The Chesapeake Bay Area Exoplanet Meeting #12, Laurel, USA | <i>May 2025</i> |
| 7. VICO Spring Workshop 2025, Charlottesville, USA | <i>May 2025</i> |
| 8. UVA Postdoc Research Symposium | <i>October 2024</i> |
| 9. Born in Fire: Eruptive Stars and Planet Formation, Santiago, Chile | <i>September 2024</i> |
| 10. New Heights in Planet Formation, Garching, Germany | <i>July 2024</i> |
| 11. ESO Wine & Cheese Seminar | <i>January 2024</i> |
| 12. European Astronomical Society Annual Meeting (EAS), Krakow, Poland | <i>July 2023</i> |
| 13. Meeting of ALMA Young Astronomers (MAYA), Online | <i>March 2023</i> |
| 14. Annual Meeting of Astronomy Society of the Republic of China (ASROC), Taipei, Taiwan | <i>September 2020</i> |

Posters:

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| 1. European Astronomical Society Annual Meeting (EAS), Cork, Ireland | <i>June 2025</i> |
| 2. Origins of Solar Systems (Gordon Research Conference), South Hadley, Massachusetts, USA | <i>June 2025</i> |
| 3. Inter+Stellar: Harnessing the Intersection Between Stars and the Interstellar Medium, Baltimore, USA | <i>May 2025</i> |
| 4. Star@Lyon conference, Lyon, France | <i>June 2023</i> |
| 5. Protostars and Planets VII, Kyoto, Japan | <i>April 2023</i> |
| 6. Annual Meeting of Astronomy Society of the Republic of China (ASROC), Taipei, Taiwan | <i>September 2020</i> |
| 7. Annual Meeting of the Physical Society of Taiwan (TPS), Pingtung, Taiwan | <i>February 2020</i> |
| 8. Exploring the Universe: Near Earth space science to extragalactic astronomy (EXPUNIV2018), Kolkata, India | <i>November 2018</i> |

PUBLICATIONS

Total citations: 301, first author citations: 112. h index = 10.

Major-Contribution Publications:

1. *(Submitted) DECO x: Evidence of widespread interactions between planet-forming disks and their surrounding clouds*

- A. Gupta**, T. J. Haworth, A. Miotello, et al..
2. *(Submitted) Disk Asymmetries Grow with Stellar Mass: Evidence from 67 Planet-forming Disks*
A. Zuleta, **A. Gupta**, P. Curone, et al.
 3. *Angular Momentum of Planet-Forming Disks: Implications for Infall Driven Misalignments*
A. Gupta, C. Longarini, L. I. Cleeves, et al. (2026), accepted in A&A. [10.48550/arXiv.2607.23741](#)
 4. *A tale of three tails: A misaligned streamer and mysterious structures around [BHB2007]-1*
A. Gupta, A. S. Hales, L. I. Cleeves, et al. (2026), ApJ, 998, 105. [10.3847/1538-4357/ae2477](#)
 5. *Discovery of an Accretion Streamer and a Slow Wide-angle Outflow around FU Orionis*
A. S. Hales, **A. Gupta**, D. Ruíz-Rodríguez, et al. (2024), ApJ, 966, 96. [10.3847/1538-4357/ad31a1](#)
 6. *TIPSY: Trajectory of Infalling Particles in Streamers around Young Stars. Dynamical analysis of streamers around S CrA and HL Tau*
A. Gupta, A. Miotello, J. P. Williams, et al. (2024), A&A, 683, A133. [10.1051/0004-6361/202348007](#)
 7. *A dusty streamer infalling onto the disk of a Class I protostar: ALMA dual-band constraints on grain properties and mass infall rate*
L. Cacciapuoti, E. Macías, **A. Gupta**, et al. (2024), A&A, 682, A61. [10.1051/0004-6361/202347486](#)
 8. *Reflections on nebulae around young stars: A systematic search for late-stage infall of material onto Class II disks*
A. Gupta, A. Miotello, C. F. Manara, et al. (2023), A&A, 670, L8. [10.1051/0004-6361/202245254](#)
 9. *Effects of Magnetic Field Orientations in Dense Cores on Gas Kinematics in Protostellar Envelopes*
A. Gupta, H.-W. Yen, P. Koch, et al. (2022), ApJ, 930, 67. [10.3847/1538-4357/ac63bc](#)
 10. *Interplay between Young Stars and Molecular Clouds in the Ophiuchus Star-forming Complex*
A. Gupta, W.-P. Chen (2022), AJ, 163, 233. [10.3847/1538-3881/ac5cc8](#)

Other Publications:

1. *JWST-DECO: The impact of accretion on mid-infrared observable water in planet-forming disks*
J. K. Calahan, T. Dziire, ... **A. Gupta**, et al. (2026), ApJ, 1004, 213. [10.3847/1538-4357/ae707a](#)
2. *The environmental imprint on molecular layering in the dusty streamer of M512*
M. De Simone, L. Cacciapuoti, ... **A. Gupta**, et al. (2026), A&A, 709, L10. [10.1051/0004-6361/202659631](#)
3. *Spatially correlated stellar accretion in the Lupus star-forming region: Evidence for ongoing infall from the interstellar medium*
A. J. Winter, M. Benisty, ... **A. Gupta**, et al. (2024), A&A, 691, A169. [10.1051/0004-6361/202452120](#)
4. *Anatomy of the Class I protostar L1489 IRS with NOEMA: disk, streamers, outflow(s) and bubbles at 3 mm*
M. Tanious, R. Le Gal, ... **A. Gupta**, et al. (2024), A&A, 687, A92. [10.1051/0004-6361/202348785](#)
5. *The VLT MUSE NFM view of outflows and externally photoevaporating discs near the Orion Bar*
T. J. Haworth, M. Reiter, ... **A. Gupta**, et al. (2023), MNRAS, 525, 4129. [10.1093/mnras/stad2581](#)
6. *The JCMT Transient Survey: Four-year summary of monitoring the submillimeter variability of protostars*
Y.-H. Lee, D. Johnstone, ... **A. Gupta**, et al. (2021), ApJ, 920, 119. [10.3847/1538-4357/ac1679](#)
7. *No impact of core-scale magnetic field, turbulence, or velocity gradient on sizes of protostellar disks in Orion A*
H.-W. Yen, B. Zhao, P. M. Koch, **A. Gupta** (2021), ApJ, 916, 97. [10.3847/1538-4357/ac0723](#)
8. *VR CCD photometry of variable stars in the globular cluster NGC 4147*
S. Lata, A. K. Pandey, ... **A. Gupta**, et al. (2019), AJ, 158, 51. [10.3847/1538-3881/ab22a6](#)